Kruvel Worku, Isaiah Jones, Tyler Potthoff, Savanna flores

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The Nebraska Department of Transportation has an Excel-based process for designing a concrete mix for one cubic yard of concrete using the Absolute Volume Method. While the Excel process works, it can be prone to errors if people are entering numbers by hand or if formulas are broken. The aim of our project was to take the same process and implement it in a Python program that would be more consistent and easier to replicate the same way each time. We accomplished this by translating the equations from the spreadsheet into Python functions and creating a step-by-step input system that asks for inputs in the same order as the submittal sheets. Once the user has entered the information, the program will compute the entire mix and spit out a summary table for one cubic yard.

It requests the concrete type, weights of cementitious materials and cement, water/cement ratio, air content, split of aggregates, and specific gravities. Then it calculates water weight using water weight equals total cement weight times water/cement ratio. Next, it converts weights to volumes using the specific gravities and a water weight of 62.4 pounds per cubic foot. The program then subtracts the volumes of cementitious materials, water, and air from 27 cubic feet to obtain the remaining volume of aggregates. It then divides the remaining volume of aggregates into fine and coarse aggregates using the desired percentage split and converts it to fine and coarse aggregates using weights. A yield check verifies that the final volume equals 27 cubic feet, indicating one cubic yard.

To validate the program, the program was tested using four different NDOT based scenarios that simulate different applications. In the first scenario, Class 47B paving, the program calculated 272.80 lb of water, 2004.12 lb of fine aggregate, and 872.02 lb of coarse aggregate. In the second scenario, Class 47BD bridge deck, the program calculated 304.00 lb of water, 1048.90 lb of fine aggregate, and 1597.37 lb of coarse aggregate. In the third scenario, Class 47BR revised or optimized, the program calculated 248.00 lb of water, 1771.89 lb of fine aggregate, and 1199.29 lb of coarse aggregate. In the fourth scenario,

Class PR3 high early strength, the program calculated 266.00 lb of water, 1273.57 lb of fine aggregate, and 1580.35 lb of coarse aggregate. The program was able to pass the yield check at 27.00 cubic feet.

The results make sense based on the inputs. The water weight, for example, is based on both the water cement ratio and the total cement weight, so that even though the water cement ratio was reduced, the cement weight was higher, and that's why the bridge deck mix had the highest water weight. The amount of air affects how much solid material can be included in one cubic yard, and that's why higher air means less space for aggregates. The fine and coarse split clearly affects the final weights, with Scenario 1 being fine heavy, while Scenarios 2 and 4 were more coarse heavy. The Python system followed the NDOT style, was consistent, and reduced errors that might occur with spreadsheets.